

What does research tell us about scouting in football? A scoping review with evidence gap-map

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Sérgio Ribeiro^{1,2}, Filipe Casanova^{3,4} , José Afonso⁵ ,
Alberto Pompeo^{3,4}, Everton Luís Rodrigues Cirillo^{4,6},
Sixto González-Víllora⁷ , Pedro Teques^{1,2} , Daniel Duarte^{1,2},
and Andrew Mark Williams⁸

Abstract

Aim: We examined scientific literature focusing on scouting practices in professional football, as well as presenting an evidence gap-map to facilitate future research.

Methods: The study followed the PRISMA 2020 and Cochrane's guidelines, and PRISMA extension for Scoping Reviews (PRISMA-ScR). The databases used were PubMed, Scopus, SPORTDiscus, and Web of Science. The eligibility criteria followed the PECOS approach.

Results: From 25,356 hits, 15 studies were eligible for inclusion in this scoping review, most were published between 2010 and 2020 (67%). Moreover, 80% of the studies were published over the last 10 years, and 33% (5 studies) since 2020. Additionally, we identified that the interest of studying this domain occurs most frequently in Europe. Moreover, the level of experience and qualifications possessed by the scouts plays a crucial role in their effectiveness. Finally, most of the studies selected pointed out the lack of academic or specific qualifications for scouts.

Conclusion: This scoping review underlined the complexity and variability of scouting practices in football, highlighting the need for standardization and the development of more robust and inclusive methodologies and instruments to better prepare scouts for the challenges of talent identification. Additionally, a lack of detailed demographic data and knowledge of the specialized qualifications possessed by scouts limits a full understanding. In this vein, structured evidence-based development programmes and the use of emerging technologies in football scouting may revolutionize the talent identification process.

Keywords

Recruiters, soccer, tactical knowledge, talent identification, technical skills

Introduction

The conceptual elements of detecting, identifying, developing, and selecting talented players,¹ all play a crucial role in football.^{2,3} In this vein, the research distinguishes between the interrelate constructs of talent detection and development as well as talent identification and selection.¹ Talent detection can be considered as the first stage of the process with the challenge being to discover children, as well as adolescents, with good athletic talent or potential to participate in sports training.^{1,2} Additionally, talent identification is the process of identifying 'talent' through the observation or measurement of individual skills or attributes to systematically explore and cultivate such characteristics (i.e., looking for young athletes who have the potential to become outstanding athletes).^{1,3} The talent

Reviewer: Nuno André Nunes (Solent University, UK)

¹N2i, Instituto Politécnico da Maia, Maia, Portugal

²Universidade da Maia, Maia, Portugal

³Lusófona University- Centro de Investigação em Desporto, Educação Física, Exercício e Saúde (CIDEFES), Portugal, Lisboa, Portugal

⁴Lusófona University- Núcleo de Estudos e Investigação em Futebol (NEIF), Lisboa, Portugal

⁵Centre for Research, Education, Innovation and Intervention in Sport (CIF2D), Faculty of Sport of the University of Porto, Porto, Portugal

⁶Sports Science Department, State University of Londrina (UEL), Brazil

⁷Sport and Physical Activity Education Research Group, Faculty of Education, University of Castilla-La Mancha, Albacete, Spain

⁸School of Sport, Health, and Exercise, Loughborough University, Loughborough, UK

Corresponding author:

Filipe Casanova, Lusófona University- Centro de Investigação em Desporto, Educação Física, Exercício e Saúde (CIDEFES), Portugal, Lisboa, Portugal; Lusófona University- Núcleo de Estudos e Investigação em Futebol (NEIF), Lisboa, Portugal.

Email: filipe.casanova@ulusofona.pt

pathway is viewed on a continuum from immediate performance (present) to predictive success (future).

In football, it is common that scouts may often identify and select promising young players by assessing their 'talent' during competition. Scouts need to make informed decisions to select the most promising players to excel at the elite level.⁴ In this context, scouts, a term used to address the talent scout or recruiter,² are responsible for the process of identifying talent to meet the needs of clubs, academies and national teams to build a solid foundation for success.⁵

The need to discover talented football players presents several challenges to scouts⁶ and its inherent complexity is partly due to the specificities of each stage of talent development (e.g., each stage plays an integral part of the talent training process).¹ Moreover, scouts are faced with the difficult task of selecting future top-level players at an early stage in their careers,⁶ and there is constant pressure to identify promising young players before other rival clubs or associations. The difficulty of early talent selection, and the related inappropriate exclusion of late-developing talent, encourages an extended and long-term focus on development.⁷ Ultimately, talent detection and identification should be seen as ongoing processes, and talent selection should be reserved for later stages and be dynamically revised since performance is inherently multifaceted and complex.³ Therefore, talent identification and its development are multidimensional and nonlinear.

In top-level elite youth football, clubs and associations are now increasingly reliant on scouts, who take on a prominent role as specialists responsible for identifying, evaluating, and recruiting young talented players.^{5,8} Such a task necessitates that scouts evaluate tactical-technical skills,⁹ as well as physical and psychological characteristics.¹⁰

Beyond football-specific skills and physical attributes, scouts can rely on their knowledge of players' specific education, years of experience and the systematic exchange of know-how with top-level colleagues.^{11,12} Potentially, differences in the scouts' qualifications and training (i.e., coaching license, academic graduation, professional experience, familiarization with different levels of practice and stages of development) may lead to poor or inconsistent identification and selection of players. Therefore, it seems that the success or failure of scouts relies on their knowledge and sport-specific background.

While scouting in football is crucial, academic literature dedicated to this topic is scarce. Most available knowledge derives from practical experience, with only a few scientific studies analysing the procedures, criteria, and tools used, or the typical profile of scouts.^{1,4} This gap in our knowledge can hamper the effectiveness of the talent detection and selection process in football, leading to an under-use of resources and possible loss of young talented players. A scoping review can provide a first step to organizing the existing literature and mapping existing knowledge about

those involved in scouting young football players (e.g., observation strategies and criteria, data collection tools, qualifications). Likewise, relevant knowledge gaps can be identified and help guide future research^{1,4} and an evidence gap map (EGM)¹³ may help to highlight the most relevant areas for investigation. This information may provide valuable guidance for scouts, coaches and other professionals interested in improving their skills to identify and develop talented football players, while at the same time, inspiring clubs, academies, associations, and other education institutions to foster structured programmes to train scouts.

Materials and methods

This scoping review with EGM was registered on the Open Science Framework (OSF) platform on May 31, 2023, three months before the searches were performed (project: <https://osf.io/hrm4q>; registration doi: <https://doi.org/10.17605/OSF.IO/7XNTF>). We followed the PRISMA 2020 guidelines,¹⁴ and PRISMA extension for Scoping Reviews (PRISMA-ScR)¹⁵ and Cochrane's guidelines.¹⁶

Eligibility criteria

We included original research papers published in peer-reviewed journals, without restrictions concerning publication date or language, and no filters applied. Eligibility criteria followed the Participants, Exposure, Comparators, Outcomes, Study Design (PECOS) approach: (1) football (soccer) scouts with no restrictions concerning sex or age; (2) exposed to processes of detection, identification, and/or selection of football players with no restrictions concerning player sex, age, or participation level corresponding to the Participant Classification Framework (PCF)¹⁷ Tier 1 – Recreationally Active, Tier 2 – Trained/Developmental, Tier 3 – Highly Trained/National Level, Tier 4 – Elite/International Level and Tier 5 – World Class, but excluding Tiers 0 (Sedentary); (3) comparators were optional; (4) any data related to the profile of the scouters (e.g., qualifications); (5) no limitation was placed regarding study design.

Information sources

Searches were conducted on September 26, 2023, and updated on November 14, 2024, using PubMed, Scopus, SPORTDiscus and Web of Science (core collections) databases. Manual research was conducted using the reference lists of the included studies. Afterwards, snowball citation tracking was performed using Web of Science. To gather additional suggestions for potentially relevant studies, two external experts (PhD, with published research on the topic) were consulted (NG and JM-A). A waiting period of three weeks was allotted for the first response, with a reminder sent after the initial two weeks. In the event of a positive response, an additional four-week period was

granted to complete the task. Finally, a search was conducted to identify any errata or retractions related to the studies included.¹⁶ If any pre-registered protocols or supplementary files existed for the included studies, these were collected.

Search strategy

The general search strategy was used of free text terms and the Boolean operators AND/OR, with one line of code referring to the Title/Abstract: (*Soccer OR Football**), and three code lines referring to all fields: (*Identif* OR Recogniz* OR Recruit* OR Select* OR Deselect* OR Detect* OR Develop* OR Predict* OR Track**) AND (*Talent* OR Gift* OR Expert**) AND (*Scout* OR HeadHunt* OR Coordinat* OR Coach* OR Decis**). The full search strategies for each database are presented in Supplementary Table 1.

Selection process

Automated removal of duplicates was performed using EndNoteWeb (ClarivateTM), but further manual removal of duplicates was required. SR and AP independently screened each record retrieved in two stages: (i) screening of titles and abstracts; (ii) full text analysis. In case of disagreements between the two authors, FC provided arbitrage until consensus was achieved.

Data collection process

SR and AP performed independent data collection from reports. In the event of disagreements between the two authors, FC provided arbitration until consensus was reached. All data were organized and coded using a specifically designed Microsoft[®] Excel worksheet. The final data on competitive level were coded by three authors (SR, AP and FC) using Tier 1 or higher of the Participant Classification Framework.¹⁷ If any relevant data or contextual information were missing, the authors of the respective studies were contacted via email, allowing a three-week period for response (including a reminder after the first two weeks). If no response was received within three weeks, and the missing information was deemed necessary based on the eligibility criteria, the study was excluded. However, if the missing information was not crucial for the eligibility criteria, the study was included in the review.

Data items

Outcomes: indicators characterizing the football scouts involved in detection, identification, and/or selection of players. This included (but was not limited to): (i) age, sex, qualifications (e.g., academic degrees, coaching licenses), profile, criteria implemented to perform their

tasks; (ii) age, sex, and competitive level of the groups where the scout acts, among others. Competitive levels were classified according to the PCF.¹⁷ Additional information included (but was not limited to): sample size, geographical location, funding, and competing interests.

Data management and synthesis methods

A narrative synthesis was executed, accompanied by data summaries (number, percentage) for the pre-defined data elements. To present an overview of the current body of knowledge and the associated research deficiencies, an EGM was built to illustrate the existing evidence landscape and the present research gaps.^{13,18–20}

Results

Study identification and selection

The search yielded a total of 25,356 hits (Figure 1). Of these, 11,786 duplicates were removed. The remaining 13,570 records were analysed for relevance based on their titles and abstracts, of which 12,669 were excluded. From the remaining 901 full texts, 684 were excluded based on not being original research papers, and a further 162 were not published in peer-reviewed journals. Finally, a total of 43 studies^{21–62} were excluded because they did not meet the PECOS criteria (i.e., Criteria 1: $n = 9$,^{23,26,29,31,37,47,48,50,61} and criteria 2: $n = 34$ ^{21,22,24,25,27,28,30,32–36,38–46,49,51–60,62}), leaving 12 studies for data extraction and further analysis.^{56,63–73}

Two more articles were identified as eligible through snowballing, which were reviewed and integrated,^{74,75} and after this list was reviewed by the external experts, one more study identified through the list of references was suggested.⁷⁶ To meet the required eligibility criteria, the authors of one study⁷⁷ were contacted by email and via ResearchGate, simultaneously. However, no response was obtained, so this work was excluded. Ultimately, 15 studies were included in the review^{63–76} (Figure 1).

Study characteristics

Characterization and contextual-related information of the included studies. The studies were published between 2009⁷⁶ and 2023.^{64,67} Most studies (67%) were published between 2010 and 2020.^{56,63,65,66,69–73,75} Altogether, 80% of the studies were published over the last 10 years, and 33% since 2020.^{56,64,67,68,74} Only one study (7%) was published before 2010.⁷⁶

From the 15 selected studies, only six provided the information about the sample size of the athletes that scouts evaluated.^{56,65,67,69,70,72} The smallest sample used was 24 athletes⁶⁷ and the largest was 556 athletes,⁵⁶ and the age of the athletes ranged from U7 to professional athletes,

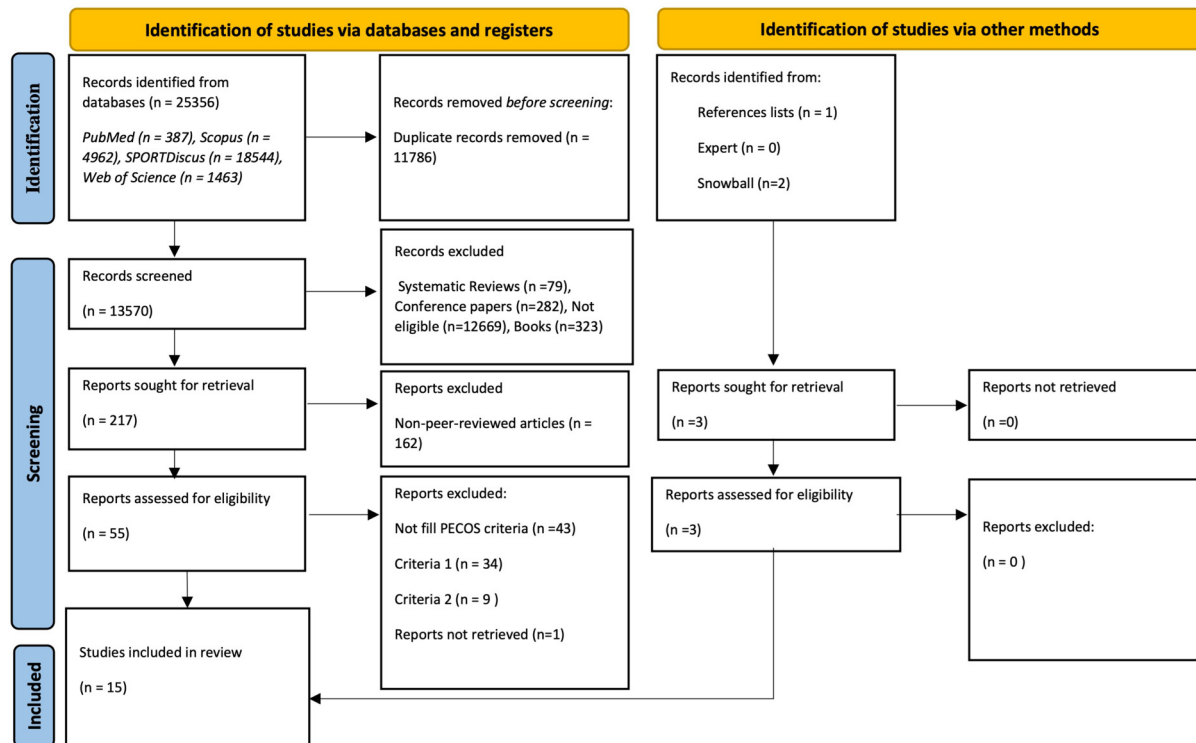


Figure 1. PRISMA 2020 flow diagram.

but three studies did not report any information^{64,68,76} (Table 1). Additionally, eight of the 15 included studies reported that the athletes who participated in the studies were classified at a Provincial/State or Academy Programs (Tier 3).^{56,65–67,69,71–73}

Procedures used by the scouts to identify talent players

Regarding the evaluation methods implemented, 10 studies used questionnaires or interviews semi-structured with the scouts.^{63,64,66,67,69,71,73–76} As for the other five studies,^{56,65,68,70,72} the methods were varied, ranging from analysis of physical, technical, tactical knowledge or using a mathematical model that prioritizes player attributes, among others.

Different methods were used to understand the scouts and coaches mindset or procedures, including questionnaires with 10,⁷³ 18⁶³ and up to 36 questions⁶⁴ employed. Other studies have used standardized questionnaires such as the Nomination Scale for Identifying Football Talent (NSIFT),⁵⁶ and Qualtrics.⁷⁴ In the study conducted by Orosz and Mezo⁶⁹ more instruments were used, where some of them were used for athletes and others for coaches, particularly; Sports Background Questionnaire (SBQ), Tennessee Self-Concept Scale (TSCS), Psychological Immunological Competence Inventory (PICI), Athletic Coping Skills

Inventory (ACSI); Advanced Progressive Matrices (APM), and specific questionnaires were administered to the players' coaches and teammates. However, to identify talented football players only a few studies^{56,65} reported that scouts or coaches used specific-football instruments (e.g. GPS, GTS, GPET) or maturational indicators (e.g., SA, age prediction by PHV).⁷² Additionally, only three studies^{65,69,75} tried to identify the performance indicators that scouts used to identify talent youth players during different time points (i.e., longitudinal study).

Moreover, Larkin and O'Connor⁷⁵ identified and evaluated the main characteristics and attributes considered important by football recruiters and coaches for identifying talent young players, categorizing into technical, tactical and psychological skills as the main criteria. However, the psychological criteria were the most frequently highlighted by scouts or coaches.^{56,63,64,66,68,69,71} Additionally, Pedroza-Junior⁷³ suggested the importance of restructuring and organizing the youth department in a professional manner, creating an exclusive department for attracting and managing young footballers. Recently, the characteristics of a talented player were categorized as personality factors, teamwork skills, decision-making, tactical-technical skills and expectations of success,⁶⁴ independently of playing position. Moreover, Ozceylan⁷⁰ proposed a solution where scouts prioritised the main attributes of each player based on the positions by using the AHP.

Table 1. Characterization and contextual-related information of the included studies.

Study	Aims	Number of Player and Gender	Player's Competitive level	Methodology	Instruments	Main Findings
Christensen ⁷⁶	- To explore how elite football coaches identify and develop talent in young players. - To examine how talent identification is influenced by social and subjective factors such as the coach's "taste" and visual experience. - To understand how coaches construct their cognitive structures and exercise their power as arbiters of talent in elite football.	- NA	- NA	- In-depth interviews with elite football coaches was used. - Phenomenological analysis known as "condensation of meaning" inspired by the phenomenological psychologist Amedeo Giorgi.	- No specific instruments used in the study were mentioned. - Data was collected through in-depth interviews, which were audio-recorded and later transcribed.	- Elite football coaches used a combination of visual observation and subjective assessment to identify talent in young players. - Talent identification is influenced by social and subjective factors, such as the coach's "taste" and visual experience. - Coaches build their cognitive structures and exercise their power as arbiters of talent in elite football. - The construction of coaches' expertise is a continuous and dynamic process involving reflection on practice and learning through experience.
Paoli et al. ⁷¹	- To describe and analyse the process of detecting and selecting football talent, considering identity representations and practices in everyday football.	- NA	- U15 - U17 - U20 - Tier 3	- Semi-structured interviews carried out with professionals directly involved in the training of football players, from the seven clubs. - The interviews were transcribed in full and analysed in triangulation with fieldwork observations and technical and socio-anthropological football literature.	- Semi-structured interviews as the main instrument for collecting data. - Fieldwork observations and technical and socio-anthropological football literature.	- The talent selection process is structured by a calculated "pedagogical project" focused on market aims, forming players as products for the market. - The training of young athletes is oriented towards a more physical style of play, valuing athletic characteristics and discipline. - Contrary to the myth of the naturally gifted Brazilian player, the evaluation and training process is continuous, labor-intensive, and long-term, with effectiveness proven

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive level	Methodology	Instruments	Main Findings
Pedroza-Junior ⁷³	- To collect, interpret and understand data relating to the selection process applied to male field football players in U15 in Pernambuco region. - To compare the results obtained in the questionnaire with the relevant scientific literature.	- NA	- U15 - Tier 3	- Structured questionnaire with 10 questions, in person, to social actors directly linked to the selection process of the Náutico Capibaribe, Sport Recife, Santa Cruz and Atlético do Porto clubs, specifically professionals linked boys' field football. - Descriptive statistics such as absolute and relative frequencies and averages were also used to analyse the data collected.	- Structured questionnaire with 10 questions.	through systematic training, assessments, and sports competitions. - There was no more or less appropriate way of identifying young talent, nor was there a consensus on the age or methodology used to do so. - The importance of restructuring and organizing the youth department in a professional manner was highlighted, creating an exclusive department for attracting and managing young football.
Brîndescu ⁶³	- To obtain expert opinions on the main aspects of selecting children aged between 6 and 8 years of football practice, including selection criteria, methods of encouragement, stimulation and improvement, and the general state of the organised selection phenomenon for children in this age group.	- NA	- U7 to U9 - Tier 2	- A questionnaire of quality of sports selection for children aged 6 to 8 in schools and sports and private clubs, and for expert opinions on key aspects of selection in this age group, such as selection criteria and methods of encouragement and improvement.	- Questionnaire consisting of 18 questions, of which 1 had a closed answer and 17 had open answers.	- The majority of experts believe that initial selection in football should be carried out between 6 and 8 years of age. - The most effective criteria for initial selection in football were motor criteria, followed by psychological criteria. - Experts' opinions vary on ways to improve initial selection in football, but most agree that organizing regional squads and trials in schools is important. - Training methods and the motivation of players are considered vital elements for the successful selection of 6–8

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive level	Methodology	Instruments	Main Findings
Goto et al. ⁶⁵	- To examine the distances and speeds travelled during football matches by U11 to U16 English Premier League Academy players. - To analyse the differences in match performance between players subsequently retained or released by the academies. - To establish the effect of calculating speed zones in different ways when analysing match performance.	- 81 - NA	- U11 to U16 - Tier 3	- Performance analysis in football matches, sprint tests, statistical analysis and the use of a Global Positional System (GPS) to monitor players' performance during football matches.	- GPS system (SPI Elite, GPSport, Australia). - Photoelectric stopwatch (Brower timing, USA). - Team AMS software version 1.2 (GPSport, Australia).	- U11 to U16 players from the English Premier League Academy covered an average of 5.6 to 7.2 km per game, with average speeds ranging from 4.8 to 5.8 km/h for walking, 9.1 to 11.1 km/h for running and 22.1 to 24.2 km/h for maximum sprints.
Orosz & Mezo ⁶⁹	- To examine the psychological factors that affect the development of talent in football. - To measure the integrative aspect of the practical validity of sports talent	- 425 - NA	- U15 (n=96) - U16 (104 players) - U17 (106 players) - U18 (41 players) - U19 (78)	- Quantitative approach in several stages was used. The players and coaches completed questionnaires and psychological tests in the presence of a psychologist. - Discriminant statistical analysis with a stepwise	- Sporting Background Questionnaire (SBQ). - Tennessee Self-Concept Scale (TSCS). - Psychological Immune Competence Inventory (PICI). - Athletic Coping Skills Inventory (ACSI).	- The most talent players showed significant differences in a variety of psychological factors compared to players considered to be less talented, these factors included: self-concept; coping skills; self-confidence and motivation;

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive Level	Methodology	Instruments	Main Findings
Jokuschies & Conzelmann ⁶⁶	development in a large sports sample, with a view to preventing talent loss and seeking talent development. - To compare the personal and interpersonal importance of players considered talented and less talented by coaches and team-mates.	- NA	players) - Tier 3	method, and talent act-tope model to integrate the results of the tests and questionnaires.	- Advanced Progressive Matrices (APM). - Specific questionnaires to assess the players' perception of talent and performance.	intelligence; better performance under pressure, more concentration and resilience to anxiety, more openness to challenges, more self-efficacy and social skills.
Ozceylan ⁷⁰	- To analyse the talent criteria of top-level football coaches. - To describe how coaches' subjective models of talent can be defined using a constructivist approach. - To analyse the similarities and differences between different coaches' models of talent and investigate whether coaches consider players based on these subjective criteria to have the potential for a professional career in football. - To present a football player selection	- 30 - NA	- U14 to U18 - Tier 3	- Approach based on the quality criteria of qualitative research, considering procedural documentation, rule-orientation and communicative validation. - A qualitative method with an inductive approach was adopted, where the construct and contrast poles for each talent criterion were determined, using an interview guide to specify the content.	- Semi-structured interviews and the repertory grid technique were used. - A constructivist approach was used to study swiss top-level youth soccer coaches using a semi-structured interview and repertory grid technique.	- Subjective talent criteria of high-level football coaches can be effectively captured through semi-structured interviews and the repertory grid technique. - The multidimensionality and differentiation of coaches' talent models was demonstrated, covering areas such as personality, cognitive and perceptual abilities, physical abilities, technique and social environment. - The repertory grid technique can be used for systematic evaluations of talent based on coaches' subjective criteria.
			- Professional team Fenerbahçe	- A football player selection approach based on a mathematical model that uses	- Analytic Hierarchy Process (AHP), and the 0–1 integer linear programming model.	- The proposed approach can be effective in selecting football players based on their

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive level	Methodology	Instruments	Main Findings
Larkin & O'Connor ⁷⁵	approach based on a mathematical model. - To identify and evaluate the main characteristics and attributes considered important by grassroots football recruiters and coaches for identifying talent in young players at the entry level of representative football. - To determine the hierarchy of attributes essential to the performance of U13 players, highlighting the importance of technical, tactical and psychological attributes in identifying talent in youth football.	- NA	- U13 - Tier 2	the analytic hierarchy process (AHP). - Individual face-to-face interviews, open coding, group discussions with a panel of three experts to group similar units of meaning, operational definition of each attribute identified, and the application of questionnaires.	- Semi-structured interviews, questionnaires used to identify attributes and importance rating scales, such as the Miller Scale, which ranged from 0 to 9 points, and the modified Delphi method was used. - The modified Delphi method was adopted, involving an initial interview followed by two rounds of questionnaires with experts in the field of youth football.	attributes and positions within the team. - In addition, it highlights the importance of using mathematical and analytical approaches in the selection of football players. - The hierarchization of the essential attributes for the performance of U13 were seven considered to be the most important and eleven to be as moderately important. - The emerging attributes were categorised into technical, tactical and psychological skills. - The importance of technical proficiency stood out, with skills such as first touch and one-on-one ability being considered crucial. - Talent selection is crucial for developing elite players, with coaches seeking key attributes for a successful career. - Researchers recommended a holistic, multidisciplinary approach to talent identification, valuing technical, tactical, and psychological attributes more than physiological, anthropometrical, and sociological ones for U13 teams.
Romann et al. ⁷²	- To investigate the validity of the "coaches' eye"	- 144	- U15	- A comparison of three approaches to assessing	- Radiographs of the hand and wrist, and subjective	- The coach's eye is a valid method for assessing biological

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive Level	Methodology	Instruments	Main Findings
Serra-Olivares et al. ⁵⁶	method in assessing biological maturation in youth football players, comparing it with more objective methods such as skeletal age and peak height-velocity (PHV). - To examine the implications of these findings for the selection and development of talent in youth football. - To propose a multidisciplinary approach for evaluating young football players' talent.	- NA - 556 - NA	- Tier 3 - U10 to U23 - Tier 3	biological maturation in young football players: Skeletal age (SA), age prediction by PHV and an estimate of biological maturation by the "coaches' eye". - Based on the dynamic ecology of movement for the contextualized analysis of the variables that determine sporting performance. - Tactical knowledge analysis, integrated assessment of physical performance, decision-making processes, and technical-tactical execution.	observations by the coaches to estimate biological maturation using the "coaches' eye" method. - Nomination Scale for Identifying Football Talent (NSIFT). - Interview through video sequences on decision-making. - GPS, accelerometry, and video monitoring in training and/or competition. - Game Test Situations (GTS) and Game Performance Evaluation Tool (GPET). - WIMU monitoring device, from Real Track Systems (Quiqo Software).	maturation. In fact, this method was even better than the widely used age of peak height velocity assessment. - Compared to the classic X-ray method, the coach's eye offers much faster information collection, lower costs, and no radiation exposure. - In sports, the systematic and widespread implementation of maturity assessments could have a significant impact on performance evaluation, selection, and training during athlete development. - The combined use of contextual tests, such as the NSIFT, TCTOF, WIMU and GPET, for a more comprehensive assessment of player's performance. - The GPET was highlighted as a tool that allows for the ecological assessment of players' decision-making and technical-tactical execution, favoring the development of autonomous players ability of making intelligent and creative decisions.
Bergkamp et al. ⁷⁴	- To explore how football scouts identify talented players. - To investigate at what	- NA	- U13 to adult players - Tier 2	- Questionnaire distributed via Qualtrics was used, consisting of 8 questions divided into three sections, including	- Digital questionnaire distributed via Qualtrics.	- Players aged 12 and under could be predict, and on average at the age of 13.6 a player had the potential to become a

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive level	Methodology	Instruments	Main Findings
	age scouts perceive they can predict a player's performance. - To identify the main attributes, they consider important for future performance and to what extent they report scoring and combining this information in a structured way.			open-ended, classification and multiple-choice questions.		professional footballer. - Early indicators of future performance are often scarce, but still advise selecting players at younger ages. - It is considered more easily observable attributes, such as technical attributes, rather specific predictors of future performance. - The assess of players' attributes was by a structured way. However, the scouts' prediction (i.e., final score) on an intuitive integration of different performance attributes is a sub-optimal strategy.
Nilsen et al. ⁶⁸	- To investigate context-specific beliefs in relation to an athlete and how knowledge of these beliefs can provide new insights into the role of the coach regarding talent identification and development. - To test the athlete talent mindset (A-TM) views of coaches to gain new insights into coaching behaviour.	- NA	- NA	- A post-positivist approach to test the athlete talent mindset (A-TM) and talent identification views of coaches. - Data collection was carried out by means of e-mail invitations and online calls published on the social networks of sports organizations.	- A modified version of Dweck's Theories of Intelligence Scale (2000). - A question about the age at which coaches believe athletic talent can be detected.	- The identification of coaches' beliefs in relation to the athlete's talent mindset (A-TM) and the age at which they believe sporting talent can be identified. - However, the differences found had small effect sizes, which may indicate limitations in the practical significance of these differences.
García & López ⁶⁴	- To establish, from an empirical perspective,	- NA	- NA	- A non-experimental, descriptive empirical research design was	- Questionnaire developed by the authors, consisting of a total of	- The characteristics of a talented player were categorized into

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Table 1. (continued)

Study	Aims	Number of Player and Gender	Player's Competitive level	Methodology	Instruments	Main Findings
Lüdin et al. ⁶⁷	the elements that condition and define a talented footballer, based on the opinion of coaches. - To identify the characteristics and skills that coaches consider most important in identifying a talented footballer.	- 24 - Male	- U11 - Tier 3	used by a questionnaire-based developed by the authors. - 36 items were covered to guarantee the rigidity of the information collected.	36 items. The first four items of the questionnaire aimed to define the respondent's profile, while the following 31 items were scored on a five-point scale (from "strongly disagree" to "strongly agree").	seven factors: physical-environmental conditioning factors, interpersonal relationships, personality factors, teamwork skills, decision-making, tactical-technical skills and expectations of success. - It was highlighted the importance of the coach as an influential and decisive figure in the development of football players, as well as the relevance of the social and sporting context and individual personality in identifying and managing talent.
	- To analyse the disagreement among scouts. - To improve selection criteria.	- 24 - Male	- U11 - Tier 3	- Scouts were divided into groups to watch match recordings and rank the players from 1 to 12. After ranking, they completed a questionnaire about their evaluation methods. - Agreement among rankings was assessed using Krippendorff's α coefficient.	- Ranking of selection priority and a questionnaire that assessed the scouts' approaches, including the use of specific attributes or general impressions in their evaluations.	- Talent scouts showed low agreement in player rankings, indicating inconsistencies in selection decisions. - Most scouts relied on intuitive approaches rather than structured methods, contributing to these variations. - Adopting a more structured and standardized approach could improve agreement among scouts, increasing the accuracy of selection decisions and highlighting the complexity of the talent evaluation process in youth football.

NA: Information not available.

Characteristics of the scouts

The studies that characterized the scouts and its context-related information per year are presented in Table 2.

We identified that the interest of studying this domain occurs more frequently in Europe, with ten countries identified (Spain = two; Switzerland = two; Turkey = one; Hungary = one; England = one; Romania = one; Norway = one; Denmark = one; Netherlands = one; Germany = one). Two studies were carried out in Brazil and one in Australia (Figure 2).

Regarding the qualifications of scouts or coaches, two studies reported academic qualifications,^{70,76} three studies presented coach qualifications,^{65,66,71} while two studies reported both coach and academic qualifications,^{56,72} one study reported a scout specific-qualification,⁶⁷ and, surprisingly, nine studies did not report any information about their qualifications.^{63,64,66,68,69,73–75}

Additionally, we observed that an average of 295 scouts were used in the studies identified, ranging from 2⁶⁵ to 2499,⁶⁸ and the median value was 21 scouts, with an inter-quartile range of 90. Regarding the age of the professionals included, only the studies of Nilsen et al.⁶⁸ and Bergkamp et al.⁷⁴ presented these data. In the study of Nilsen et al.⁶⁸ most of the scouts were men and women between the ages of 16 and 83 years old. The average length of scouts' experience was ten years, with the shortest being 4.8 years,⁶⁷ and the longest being 28 years.⁷⁶ Probably, this last item supported the importance of experience in football to identify and select talented youth players, combined with their coaching practice.⁷⁶

The scouts were mainly football coaches, whereas in some studies they were coaches at youth level and others at senior level, either for participation in regional or national competitions, and some of them held a UEFA-A or UEFA-B license.^{56,65,66,72} Four studies did not report the level or age at which the scouts worked,^{64,68,70,76} and this detail was reported in the other studies, which ranged from T2 to T4.¹⁷

Evidence gap-map of the included studies

Figure 3 presents the EGM that synthesizes the patterns and gaps that were previously identified, suggesting that the order of importance for each criteria used to identify talented football players were psychological, physical, technical, experience of the scouts, tactical and subjective (e.g., scout's vision). In addition, the studies included pointed out the lack of specific qualifications possessed by those responsible for identifying and selecting future players.

Discussion

We examined the existing scientific literature that deal with scouting practices in football to promote increased

understanding of the practices used in talent detection and selection, as well as providing an EGM about the studies conducted in football. The results obtained from this review highlight the identity and qualifications of the scouts, and the strategies used to identify and select talented football players, providing an insight into the scouting process.

In the present review 15 articles were included. Altogether, four studies did not state the age of the players assessed,^{65,68,75,76} while the remaining articles assessed age groups ranging from U8 to seniors. It is important to evaluate different age groups since the characteristics and development of players vary significantly across age.² Moreover, as evidenced by EGM (Figure 3), the scouts need to establish a consensus in prioritising the main characteristics in identifying and selecting talented players for each specific-position. In this sense, the early identification of football players (6–8 years old) might focus on specific-motor and psychological criteria, and a more systematic and organized participation of the local authorities (e.g., schools, clubs) in promoting organized football competitions for children in this age group and involving families in maintaining children's interest in sports activities. Therefore, including a wide range of age groups, for example in longitudinal studies, could allow for a more complete and diverse view of scouting practices and the characteristics of players at different stages of physical and cognitive development.

Additionally, only five studies^{56,65,69,70,72} presented the sample size of the athletes evaluated, with an average of 247 athletes, ranging from 30⁷⁰ to 536.⁵⁶ This variation suggests that studies on scouting can focus either on detailed analyses of smaller groups or on broader, more generalizable studies. In this vein, Sagar⁷⁸ supports the need for a diversity of sample sizes to capture different aspects of performance (e.g., physical, tactical-technical, psychological) and their potential (e.g., quantitative and/or qualitative data). For example, Larkin and O'Connor⁷⁵ recommended a holistic, multidisciplinary approach to talent identification, valuing technical, tactical, and psychological attributes more than physiological, anthropometrical, and sociological ones for U13 teams. Moreover, the concepts of prediction and selection are inherently linked, since predictions can only be made through consideration of existing qualities for performance, and consequently attributes which contribute to current performance levels are not the only contributors for prediction. On the other hand, to strengthen the studies in this domain a wide range of age groups should be employed, and a longitudinal observational and evaluative framework is needed, reducing a gap in existing literature (Figure 3).

No consistent or standardized criteria were employed with measures covering a wide range of factors that can influence performance and future potential were applied in the most of studies selected.^{56,63,64,66,67,69,71,73,74} Albeit

Table 2. Characteristics of the scouts and its context-related information in the included studies.

Study	Country	Years of experience	Sex (Age)	Sample Size	Scouts Qualifications
Christensen ⁷⁶	Denmark	8–28 (mean = 15) years	Undefined, but the authors described that most of them were male.	8	Bachelor's degree
Paoli et al. ⁷¹	Brazil	NA	NA	38	Coaches of the U15 (n = NA), U17 (n = NA) and U20 (n = NA), and technical observers (n = 38) Qualifications or academic background NA
Pedroza-Junior ⁷³	Brazil	NA	NA	10	Professionals linked to U15 team Qualifications or academic background NA
Brîndescu ⁶³	Romania	NA	NA	100	All of them were teachers or football coaches Qualifications or academic background NA
Goto et al. ⁶⁵	England	NA	NA	2	UEFA-A license (n = 1) UEFA-B license (n = 1)
Orosz & Mezo ⁶⁹	Hungary	NA	Male	21	NA
Jokuschies & Conzelmann ⁶⁶	Germany	NA	Male	6	All of them were football coaches with UEFA-A license Academic background NA
Ozceylan ⁷⁰	Turkey	NA	NA	NA	All of them are academicians from the Physical Education and Sports School at the Gaziantep University
Larkin & O'Connor ⁷⁵	Australia	NA	NA	20	Football technical directors (n = 8) Head football coaches (n = 12) Qualifications or academic background NA
Romann et al. ⁷²	Switzerland	5 years	Male	12	Sports Sciences graduation (n = 6) National coaches of U15 to U21 teams (n = 6)
Serra-Olivares et al. ⁵⁶	Spain	≥ 10 years	Male	16	Physical Activity and Sports Sciences (n = 8) UEFA-A license (n = 8)
Bergkamp et al. ⁷⁴	Netherlands	11.2 years	Undefined, but most of the scouts were male and had an average age of 58.2 years.	125	NA
Nilsen et al. ⁶⁸	Norway	NA	Undefined, but most of the scouts were men and women between the ages of 16 and 83 years old.	2499	NA
García & López ⁶⁴	Spain	8.6 years	Male and female	873 (male) 17 (female)	NA
Lüdin et al. ⁶⁷	Switzerland	≥ 4,8 years	NA	100	The scouts took part in scouting training program provided by the Swiss Football Association (SFA).

NA: Information not available.

some efforts were detected in some studies, which ranged from analysis of technical, tactical and physical abilities to analyses using mathematical models that prioritize

player attributes.^{65,68,70,72,75,76} For example, the instruments varied widely from questionnaires with ten,⁷³ 18⁶³ and up to 36 questions⁶⁴ developed by the authors



Figure 2. Distribution of the included studies per scouts and its qualifications, continent, and country.

themselves, and some of them without specific validation for their proper use. In this sense, the lack of standardization in methods and instruments might reflect the complexity and subjectivity of scouting in football, or the need to employ systematic instruments to ensure the reliability and validity of assessments. Additionally, the instinct of scouts⁷² derives from idiosyncratic combinations of and compensations for individual performance factors (e.g., psychological, physical and technical skills, and game intelligence of the payers). Therefore, the use of a multidimensional approach,^{75,79,80} combined with a machine-learning approach⁸¹ might be crucial in developing new knowledge in this domain.

In the present study an average of 268 scouts were observed in different studies published, with some of them reporting a minimum of two⁶⁵ and a maximum of 2499 participants.⁶⁸ This large variation of the number of scouts used in the selected studies highlight considerable diversity in the methods used since it is recommended that when a small sample of scouts use the qualitative method allows a more detailed and in-depth analysis of the practices and challenges faced⁸² and with larger samples the use of quantitative methods provides a more comprehensive and statistically robust view of general trends and patterns in scouting.⁸³ Moreover, only two studies^{64,74} included information on the age of the scouts, with an average of 53 years, ranging from 33 to 64 years old. In addition, the length of experience of the scouts was highlighted, with an average of 10 years, ranging from five years⁷² to 28 years.⁷⁴ The experience characteristic and high-average age of scouts is sought to be the only certificate to detect and select players.^{84,85}

In this vein, our findings suggest that when the criteria of talent scouts were only their age or experience the results shows low agreement with player rankings, inconsistencies in selection decisions, relied on intuitive approaches rather than structured methods, personal heuristics and distinct learning experiences. Therefore, alternatively the use of mathematical models, emerging technologies (e.g., artificial intelligence, big data software) and better articulation between the football stakeholders might help in prioritizing the best player's attributes into the scouting process and future decisions. For example, Altmann et al.⁸¹ provided useful insights on which parameters might be worth monitoring with regards to talent selection processes within highly trained youth football players and how to employ supervised gradient boosting algorithms adapting the machine learning approach to a new practice context (i.e., competitive level, academies).

A geographical concentration of research in regions that also have a strong football tradition, such as Europe, was observed in the present scoping review. Of the 15 studies eligible for this review, 12 were conducted in Europe,^{56,63–70,72,74,76} two in Brazil^{71,73} and one in Australia.⁷⁵ The lack of demographic data in many studies represents a significant gap in the literature, which could compromise a better understanding on how experience and maturity influence scouting practices, as well as whether there are any regional issues that may favour particularities in the way scouts carry out their mission more comprehensive demographic information. Therefore, more studies from different countries and continents are needed

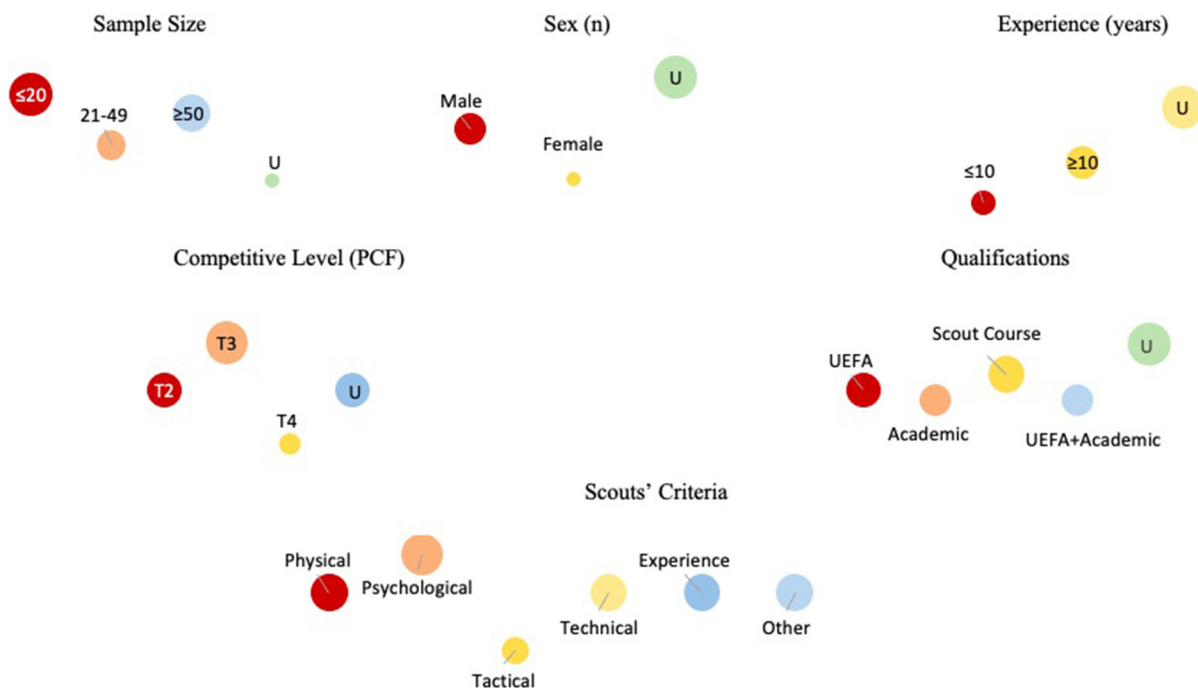


Figure 3. Evidence gap-map of scouting research.

to better understand the scouting practices according to their cultural and specific-context diversity.^{2,86}

The results of the present study revealed that scouts were mainly football coaches at youth level or adult level, holding either a UEFA-A or UEFA-B license.^{56,65,66,71,72} Only one study reported that scouts had a scouting training program provided by their national association.⁶⁷ Unfortunately, most of the studies did not report the qualifications of the scouts.^{63,64,68,69,73–75} In this sense, better qualifications (i.e., academic, sport or specific qualification) might help them to better recruit talent players and optimize the selection procedures. Moreover, the organizations (e.g., football associations, universities, clubs) should develop or adopt a structured and standardized worldwide curricular program (course) that could improve the specific knowledge among the scouts, increasing the accuracy of selection decisions and highlighting the complexity of the talent evaluation process in youth football. A good example of worldwide best practices is the talent scout education developed by Swiss Football Association and the FA talent ID pathways designed to identify and develop talented football players in England. However, more educational programs and workshops based on scientific evidence in different world regions can improve scouts' skills and knowledge, helping them to keep up to date with best practices and innovations in the field.

The results revealed a significant variation in sample size and a lack of comprehensive demographic data. To advance the understanding of football scouting, it is crucial to integrate diverse methodological approaches and increase the collection of demographic data. Professionalization and

continuous training of scouts are essential to meet current and future challenges in football. In future, to better understand the profile of the scouts, researchers should include more demographic and certification data (age, length of experience, specialized or generic formation), and stakeholders policies implemented. In addition, the use of mixed methodologies (quantitative and qualitative) should be encouraged to capture a more complete view of scouting. The creation of collaborative networks between researchers and institutions can standardize data collection and facilitate large-scale studies. The integration of qualitative and quantitative approaches can provide a richer and more complete understanding of scouting in football. In future, researchers should seek to combine these methods to capture both the subjective and objective aspects of the role.

Limitations

This study in football scouting has several limitations that should be considered. First, there is great variability in the size of the samples analysed, with an average of 268 scouts, ranging from a minimum of two⁶⁵ to a maximum of 2499⁶⁸ participants. This large variation can make it difficult to generalize the results, as studies with very small samples may not be representative of the scouting population. In addition, the lack of comprehensive social and demographic data limits the understanding of how factors such as age, gender and professional background influence scouting practices. Another limitation was that of the 15 articles analysed 12 were carried out in Europe,^{56,63–70,72,74,76} as this

geographical concentration may not reflect scouting practices in other regions around the world.

Although some researchers have used a variety of instruments to assess different abilities of players, most have not adopted a multidimensional approach. Talent assessment in football is complex and multifaceted, and a more holistic approach could provide more complete insights. It is important that in future research on scouting efforts are made to increase the size and representativeness of the samples, including more demographic data from the scouts and clearly state the age groups assessed and the qualifications of the scouts. In addition, it is necessary to broaden the geographical diversity of studies, establish standardized assessment methods, used reliable and consistent assessment instruments, and adopt a multidimensional approach to capture the complexity of talent identification in football. These recommendations can help direct future research and improve understanding of scouting practices, contributing to the development of more robust and effective methods.

The implications of this study are vast and significant, ranging from improving scouting practices and training to influencing policies for developing young athletes and future research directions. By addressing the limitations identified and implementing the recommendations suggested, clubs, organizations and researchers can advance the identification and development of talent in football, contributing to the growth and professionalization of the sport.

Conclusions

This study on football scouting offers several important conclusions that could impact both practice and future research. First, the study reveals that there was no clear consensus on the best scouting practices, highlighting the need to develop more uniform methodologies and similar instruments with the integration of different stakeholders and the creation of specific programmes by universities and national football associations. Second, the average age, length of scouts' experience and their qualifications play a crucial role in scouting effectiveness to better understand the attributes required for success in football, but the lack of more detailed demographic data and specialized qualification limits a full understanding of this influence. In addition, the lack of comprehensive demographic data on scouts and the geographical concentration of studies in Europe highlight significant gaps in the existing literature and prevent a more global and inclusive view of scouting practices. Lastly, structured evidence-based development programs are essential for the technical and personal growth of young players, where the use of mathematical models and emerging technologies in football scouting may have significant potential to revolutionize talent identification.

In summary, this study underlines the complexity and variability of scouting practices in football, highlighting the need for standardization and the development of more robust and inclusive methodologies. The results of this review provide valuable guidance for clubs, academies, associations, coaches, scouts and sports professionals interested in improving their strategies for identifying and developing talented football players, relying on multidisciplinary collaboration and more inclusive research practices.

Availability of data, code, and other materials

The original protocol and any materials not submitted with this manuscript can be consulted in the OSF project page: <https://osf.io/hrm4q>.

Contribution of each author

SR: research concept and study design, data collection, data analysis and interpretation, and writing the manuscript;

FC: research concept and study design, data analysis and interpretation, writing the manuscript, and final approval of the article to be published;

JA: research concept and study design, data analysis and interpretation, writing the manuscript, and final approval of the article to be published;

AP: writing the manuscript, data collection and data analysis and interpretation;

ELRC: data collection and data analysis and interpretation;

SG-V: data collection and data analysis and interpretation;

PT: writing the manuscript, and reviewing/editing a draft of the manuscript;

DD: writing the manuscript, and reviewing/editing a draft of the manuscript;

AMW: research concept and study design, writing and editing the manuscript.

Data availability statement

The data research is available in the paper.

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ORCID iDs

Filipe Casanova  <https://orcid.org/0000-0002-9696-9355>

José Afonso  <https://orcid.org/0000-0002-2038-393X>

Sixto González-Víllora  <https://orcid.org/0000-0003-2473-5223>

Pedro Teques  <https://orcid.org/0000-0003-4338-553X>

Supplemental material

Supplemental material for this article is available online.

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